

AQIL YUSRI

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PROFESSIONAL SUMMARY

Technology driven physics undergraduate with hands-on experience in circuit design, simulation, and hardware prototyping, including CMOS design (NGSpice, KLayout), embedded systems (ESP32, Arduino, Raspberry Pi), and PCB development (KiCad). Strong foundation in problem solving, technical analysis, and cross-functional teamwork, gained through device characterization projects, and coordinating international programmes. Seeking an Engineering Internship to apply technical and analytical skills across hardware, electronics, or system-focused roles.

EDUCATION

UNIVERSITI SAINS MALAYSIA (USM)

2023 - 2027

Bachelor of Science (Physics) with Honours

- CGPA: 3.69
- Wakalah Paynet-Yayasan Khazanah Scholarship

UNIVERSITI TEKNOLOGI MARA (UITM)

2022 - 2023

Foundation in Science

- CGPA: 3.89

PROJECTS

OTDR-Based D-Shaped Optical Fiber Temperature Sensor

- Fabricated D-shaped optical fiber probes via mechanical side-polishing as a structural modification for temperature sensing.
- Designed OTDR-based measurement setup with hotplate-controlled testing in aqueous medium.
- Analyzed backscattered signal attenuation for performance evaluation against external temperature variations.

CMOS Standard Cell Design & Simulation

- Simulated CMOS inverter in NGSpice; extracted noise margins and propagation delay.
- Simulated transmission gates in NGSpice; analyzed threshold voltage drop and implemented level restoration.
- Laid out DRC-clean NAND2 and NOR2 standard cells in KLayout using Euler path methodology.
- Characterized short-channel effects (DIBL, velocity saturation) across 0.15 μm to 10 μm channel lengths in NGSpice.

EXPERIENCE

UNIVERSITI SAINS MALAYSIA, PENANG

Feb 2025 - Mar 2025

Internship

- Competently coordinated orientation programmes for ~200 incoming international students, ensuring seamless registration, communication, scheduling and on-site support within 2 consecutive months.
- Led an exchange program involving students from Nihon Fukushi University, Japan.
- Supported administrative operations, including emailing incoming international students, assisting officers with passport and visa documentation, photographing orientation week activities, improving administrative efficiency by 40%.

CERTIFICATIONS

- Semiconductor Devices Courses, Dreamcatcher (2025)
- Advanced Semiconductor Process Technologies, Dreamcatcher (2025)
- Semiconductor Crystal Growth and Substrate Preparation, Dreamcatcher (2025)
- JLPT N3, Japan Foundation (2026)

EXTRACURRICULAR

- Member, USM Buddies — Universiti Sains Malaysia (2023 – 2026)
- Volunteer, Penang International Science Festival (PISF) (2025)

SKILLS

Software & Tools: NGSpice, KLayout (DRC / LVS), XScem, Verilog, Librelane, SkyWater 130 nm PDK, KiCad, Soldering, OTDR, ESP32, Arduino, Raspberry Pi, Python, C++, Git, Linux, Docker

Technical Skills: Material Planning, Semiconductor Devices & Microelectronics, Monte Carlo / Corner Analysis, Circuit Design & Analysis, PCB Design & Prototyping, Circuit Testing & Troubleshooting, Document Management, Event Management, Technical Reporting & Documentation

LANGUAGES

- **Malay:** Native Proficiency
- **English:** Full Professional Proficiency
- **Japanese:** Limited Proficiency (JLPT N3)
- **Mandarin:** Beginner